

ICAIMT - AI Assistant for Wealth Management «ECO InvestMind AI»

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Abstract: Managing personal finances remains a significant challenge in developing economies, where access to financial advisory services is limited and financial literacy is often low. In response to this, we propose «ECO InvestMind AI», an intelligent mobile application designed to democratize wealth management through the integration of advanced machine learning (ML) and natural language processing (NLP) techniques. Our system leverages a dual-stream hybrid architecture combining a fine-tuned BERT model for sentiment analysis of financial news and an MLP-based time series model for stock price forecasting. The application provides personalized stock recommendations by analyzing market trends, ESG scores, and sentiment signals, offering users real-time, behaviorally adaptive financial guidance. To ensure scalability and maintainability, the app is built using Clean Architecture principles and Kotlin Multiplatform, enabling seamless cross-platform performance. A relational database structure supports the integration of diverse datasets, including historical stock prices, ESG metrics, and sentiment-labeled news. Evaluation results demonstrate high predictive accuracy across multiple Kazakhstani companies and indices. This paper illustrates how AI-powered tools like «ECO InvestMind AI» can bridge the gap between traditional investment tools and modern financial advisory systems, empowering users with actionable insights and promoting inclusive participation in financial markets. The proposed hybrid algorithm achieved a total system accuracy of 96.87%, confirming the effectiveness of combining sentiment analysis and time series forecasting for intelligent investment decision-making.

Keywords: Wealth management, sentiment analysis, stock forecasting, BERT, MLP, Kotlin Multiplatform, ESG, mobile finance, Kazakhstan

1. Introduction: Financial Management Challenges in Kazakhstan

Managing personal finances remains a complex challenge for many individuals, particularly in developing economies. Professional financial advisors can be expensive, rendering their services inaccessible to much of the population. This issue is especially pronounced in Kazakhstan, where, despite notable economic growth, financial literacy levels remain relatively low. A recent study reported that nearly 30% of Kazakhstani adults with limited financial knowledge have never

invested, limiting their ability to accumulate long-term financial security and resilience (Bayakhmetova, Kadyrzhanov, & Aitzhanova, 2023).

Many people, especially young people, focus on building their careers, pursuing education, or taking care of their health, yet they often overlook the importance of managing money wisely. Lack of financial knowledge can lead to poor budgeting, mounting debt, or even falling victim to fraudulent schemes. For example, if someone does not have a clear understanding of their income and expenses, they may end up in a difficult situation that affects not only their personal finances but also the broader economy.

At the same time, many Kazakhstani people continue to rely on traditional bank deposits as their primary savings strategy. By December 2024, total individual deposits in Kazakhstani banks had reached 24.5 trillion tenge, a 23.4% increase compared to the previous year (Zakon.kz, 2024). While such deposits offer a sense of financial stability, they often fail to outpace inflation.

As noted by Timur Turlov, a prominent Kazakhstani entrepreneur, inflation continues to exert a strong influence on the country's economy, leading to the steady depreciation of money. He argues that in a period of persistent inflation, relying on low-yield instruments such as government bonds or bank deposits is a weak strategy. "Those who seek maximum peace of mind may become the most vulnerable" he warns, highlighting that such approaches could result in a gradual loss of purchasing power (Turlov, 2025).

Faced with these limitations, individuals are turning to digital platforms to diversify and grow their savings. In recent years, investment applications such as Tabys, Freedom24, Jusan Invest, and Halyk Invest have broadened market access, offering lower entry thresholds and mobile-first experiences. However, these platforms often provide only basic functionality—such as asset listings or price charts—and lack personalization or adaptive guidance. Moreover, few utilize machine learning (ML) or sentiment analysis, despite growing evidence that market sentiment significantly influences investor behavior.

In this article, we will explore how modern mobile technologies and ML algorithms are transforming traditional approaches to asset management. We begin by examining the current challenges in financial services and then show how innovative solutions can make financial planning more effective and easier to understand for a wide range of users, exploring how ML techniques can enhance wealth management applications by addressing the personalization gap in existing platforms.

Also, we introduce «ECO InvestMind AI», a mobile application designed to deliver real-time, with a new approach of personalized investment recommendation. By incorporating sentiment analysis of financial news and predictive modeling, the platform adapts to users' financial goals, behavioral patterns, and emotional contexts. Ultimately, this paper demonstrates how AI-driven tools can bridge the financial literacy gap, promote more inclusive participation in financial markets, and empower users to manage their economic future more effectively.

The main objectives of this study are:

- To design and develop a mobile application, «ECO InvestMind AI», that integrates ML algorithms for stock price forecasting and sentiment analysis of financial news.
- To bridge the gap between traditional financial advisory services and modern AI-driven digital platforms by providing real-time, personalized investment recommendations.
- To promote financial literacy and informed investment behavior among users, particularly in developing economies with limited access to financial advisory services.
- To evaluate the effectiveness of the proposed system in delivering accurate stock predictions and actionable financial insights through a user-centric mobile interface.

To address the gap between existing AI-driven financial platforms and the need for more personalized, real-time investment guidance, this study explores the development of a hybrid mobile application that integrates financial news sentiment and stock price prediction. In line with this objective, the research is guided by the following questions:

1. Can financial sentiment analysis improve prediction of stock behavior?

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2. Is it possible to build a personalized, real-time recommender system using AI?
 3. How accurate and practical is the system under real-world constraints?
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2. Literature Review of Existing AI-based Financial Systems

2.1 The Role of Financial Literacy and Behavior in Investment Decision-Making

Effective personal finance management depends on more than just access to investment tools; it requires an ability to understand, evaluate, and act upon financial information. In Kazakhstan, low financial literacy has emerged as a critical barrier to participation in investment markets. Bayakhmetova et al. highlights that individuals with poor financial understanding are significantly less likely to invest, regardless of income or age (Bayakhmetova, Kadyrzhanov, & Aitzhanova, 2023).

Compounding this issue is the proliferation of information in digital media. Inexperienced users, overwhelmed by conflicting news and social media signals, often make emotionally driven decisions—such as panic-selling during market volatility or abandoning long-term investment plans prematurely. Financial education alone may not be sufficient to counteract these behaviors unless accompanied by real-time, context-aware decision support.

Research in personal finance emphasizes the behavioral dimension of money management. Zhigitbek argues that building wealth begins with disciplined habits, such as budgeting, tracking expenses, and consistently saving a portion of one's income (Zhigitbek, 2022). Samuel in his widely cited work *The Richest Man in Babylon*, similarly, underscores the importance of setting aside at least 10% of earnings, regardless of income level. These principles point to a critical insight: financial technologies should not only offer market access but also reinforce sound decision-making behavior (Samuel, 2021).

A user-friendly interface, straightforward asset selection, and clear investment mechanisms enable beginners to take their first steps in capital management without unnecessary complications. This is especially important given the low level of financial literacy observed in many regions of Kazakhstan. In such areas, people often fall victim to financial pyramids while trying to invest and grow their capital. To help address this issue, the Astana International Exchange (AIX), with support from the National Bank of Kazakhstan, launched the Tabys app—a social initiative aimed at improving financial literacy and expanding access to investment tools. This app aims to improve investment accessibility and financial literacy in Kazakhstan. (GlobeNewswire, 2025) It allows users to start investing from as little as \$1, offering access to various securities, including exchange-traded notes and gold coins issued by the National Bank of Kazakhstan. Tabys also features educational materials to guide novice investors (Tabys App – Official Website, n.d.).

2.2 Existing Digital Platforms in Kazakhstan

At this stage, investors can utilize mobile applications provided by banks and investment platforms such as Freedom Broker, Jusan Invest, and Halyk Invest, which allow convenient trading of local stocks and bonds. These applications enable users to open brokerage accounts directly from their smartphones, eliminating the need for physical office visits.

For instance, financial management tools like CoinKeeper enable users to track their daily budgets and expenses in real time. This application provides visual tools for creating a financial plan, saving money, and managing expenses efficiently (CoinKeeper, 2023).

Freedom24, launched in 2020 by Freedom Finance, provides users with access to over 1 million financial instruments, including stocks, ETFs, bonds, and options from global exchanges. The platform offers professional market analysis and investment ideas, aiming to assist users in making informed decisions (Freedom24, n.d.).

Jusan Invest, part of the Jusan Super App, offers a comprehensive suite of financial services, including the

ability to open brokerage accounts, purchase and sell shares, and access analytics and investment ideas. The app integrates banking, shopping, insurance, and mobile communications, providing holistic financial management experience (Jusan Invest, n.d.).

Another mobile application, Halyk Invest, enables online trading of equities and bonds of leading Kazakhstani and international companies. The app provides users with access to ETFs, market news, and analytics from Halyk Finance's research team, facilitating informed investment decisions (Halyk Invest, n.d.).

While these platforms have democratized access to financial markets, especially for young or first-time investors, they often lack mechanisms for real-time personalization or intelligent financial recommendations. Investment suggestions, if present, tend to be static or based on simplistic portfolio allocation rules. Moreover, sentiment analysis—a method of interpreting the emotional tone of financial news and social media—is absent in almost all Kazakhstani apps, despite its potential to contextualize market risks and investor behavior.

Despite their progress, most local platforms lack mechanisms for real-time personalization or intelligent financial recommendations. Investment suggestions, if present, tend to be static or based on simplistic portfolio allocation rules. Moreover, sentiment analysis—a method of interpreting the emotional tone of financial news and social media—is absent in almost all Kazakhstani apps, despite its potential to contextualize market risks and investor behavior.

While CoinKeeper focuses on budgeting and Tabys emphasizes accessibility, none of these platforms fully integrate ML or LLM-based reasoning to support decision-making. This leaves a clear opportunity for more advanced, AI-enhanced applications that can help users interpret complex market signals and align investment strategies with personal goals.

2.3 AI-driven Financial Platforms Overview and Comparison

Recent advancements in AI and ML have significantly reshaped the financial technology landscape, particularly in the domain of automated wealth management and digital financial advisory services. Several leading AI-driven platforms have emerged, each applying cutting-edge techniques such as large language models (LLMs), retrieval-augmented generation (RAG), and deep learning architectures to enhance market analysis, predictive modeling, and portfolio recommendations. Among the most notable of these are AlphaFin (Li et al., 2024), FinRobot (Zhou et al., 2024), and FinGPT (Yang, Liu, & Wang, 2023).

A detailed comparison of these platforms is presented in Figure 1, highlighting their core features, strengths, limitations, and areas for improvement.

AlphaFin stands out for its innovative integration of retrieval-augmented generation (RAG) techniques with Chain-of-Thought (CoT) reasoning, allowing it to deliver not only stock predictions but also logically justified investment explanations (Li et al., 2024). Its hybrid architecture combining real-time market ingestion with deep reasoning models represents a significant advancement in AI transparency. However, this sophistication results in high computational costs, and its narrow focus on stock prediction, rather than comprehensive wealth management, limits its practical applicability.

FinRobot introduces a flexible, modular architecture that distributes complex financial tasks across specialized AI agents (Zhou et al., 2024). This multi-agent approach enhances versatility, allowing the platform to address a wide range of analytical challenges—from equity research to financial document processing. Nevertheless, the system's complexity increases orchestration overhead, and its adaptability to diverse global markets remains an ongoing challenge that requires further customization.

Feature	AlphaFin	FinRobot	FinGPT
Main Strength	Integrates RAG with Chain-of-Thought (CoT) for stock prediction and explainable reasoning.	Modular multi-agent system for equity research, document analysis, and risk assessment.	Fine-tuned LLM for financial text analysis, processing financial news and sentiment.
Architectural Approach	Hybrid RAG + CoT + real-time data ingestion.	Multi-layer architecture: Agents, LLM Algorithms, LLMops/DataOps, Multi-source LLM Models.	Single LLM fine-tuned for domain-specific financial tasks.
Explainability	High: CoT pathways allow logical justification for predictions.	Moderate: CoT-based agent decomposition improves interpretability but can be complex.	Limited: Primarily focused on sentiment and news summarization without deep decision explanations.
Adaptability to Real-Time Data	Strong: Real-time market data integration.	Moderate: Requires additional validation layers for dynamic markets.	Weak: Primarily trained on static news and reports; limited real-time capabilities.
Personalization	Limited: Primarily stock-focused; lacks full wealth management strategies.	Partial: Some adaptability across financial tasks, but not customized per user profile.	Very limited: No user-specific portfolio diversification or risk profile adaptation.
Computational Efficiency	High cost: Real-time RAG is resource-intensive.	Moderate to high cost: Multi-agent orchestration leads to processing overheads.	Relatively efficient: Text-only financial data requires less compute than real-time quantitative integration.
Regional/Global Adaptability	Limited: Focused mainly on US stock market datasets.	Moderate: Regional customization is difficult but partially possible.	Limited: Mainly focused on English-language and US-centric financial content.
Main Gaps	No full wealth management, high compute cost, biases in datasets.	Complex system management, computational inefficiency, regional data limitations.	Lack of quantitative analysis, no real-time trading strategies, minimal personalization.

Figure 1. Comparative Overview of State-of-the-Art AI-Driven Financial Platforms

FinGPT, by contrast, adopts a streamlined model specializing in financial text analysis (Yang, Liu, & Wang, 2023). By fine-tuning a large language model on financial news and corporate reports, it achieves high efficiency in extracting sentiment and summarizing financial events. While its lightweight architecture ensures cost-effective deployment, FinGPT's capabilities are limited with regard to real-time market reactivity, portfolio management, and user-centric investment advice.

Collectively, these platforms demonstrate the impressive potential of AI in financial services while also revealing persistent challenges related to personalization, real-time adaptability, and system explainability. Addressing these limitations is crucial for the next generation of financial advisory applications. In this context, the proposed ECO InvestMind AI solution is specifically designed to offer real-time, behaviorally adaptive, and personalized investment assistance, aiming to close the identified gaps and broaden inclusive access to intelligent wealth management.

The comparison presented in Figure 2 outlines the evolution of optimization techniques, metaheuristic innovations, and model interpretability approaches across a range of application areas. Beginning with the seminal work of Kennedy and Eberhart (1995), the introduction of Particle Swarm Optimization (PSO) marked an important milestone in computational optimization. Their method, inspired by social behaviors such as bird flocking, provided a simple yet highly adaptable framework that became the foundation for many later enhancements in both theoretical and applied research.

Subsequent studies expanded on these ideas to address more complex, real-world problems. Mahmoudi and Feylizadeh (2018) proposed a hybridization of PSO with the Imperialist Competitive Algorithm (ICA) to improve resource-constrained project scheduling. This approach demonstrated that combining metaheuristic techniques could offer better performance, although at the cost of increased computational effort. Similarly, Bao et al. (2020) extended PSO by integrating game theory, using Nash equilibrium principles to support multi-round decision-making processes in procurement systems. Their work emphasized the importance of fairness and sustainability in optimization under competitive conditions.

In parallel, other researchers explored broader uses of metaheuristics. Dökeroğlu, Deniz, and Kızılöz (2022) conducted a comprehensive review of recent metaheuristic algorithms for feature selection, highlighting how methods such as PSO, genetic algorithms, and newer swarm-based models are increasingly applied to high-dimensional ML problems. On the other hand, Louhichi et al. (2023) shifted the focus toward interpretability by applying Shapley Values to ML clustering tasks, addressing the growing need for transparency in complex AI models. Finally, Altybayeva (2023) proposed a novel hybrid model that combines PSO with game theory for

decision-making optimization in project management, showing how multidisciplinary integration can offer new opportunities in business and engineering contexts.

Feature	Kennedy & Eberhart (1995)	Mahmoudi et al. (2018)	Bao et al. (2020)	Dökeröglu et al. (2022)	Louhichi et al. (2023)	Altybayeva (2023)
Study Reference	"Particle Swarm Optimization" Proceedings of ICNN'95 - International Conference on Neural Networks, Perth, WA, Australia, 1995, pp. 1942-1948.	"A novel algorithm for solving resource-constrained project scheduling problems: a case study" Journal of Advances in Management, Vol. 15.	"A Novel Particle Swarm Optimization Approach to Support Decision Making in Multi Round Auction by Game Theory" International Journal of Computational Intelligence Systems, Vol. 13(1), 1447-1463.	"A Comprehensive Survey on Recent Metaheuristics for Feature Selection" Neurocomputing, Vol. 449, 269-296.	"Shapley Values for Explaining the Black Box Nature of Machine Learning Model Clustering" Procedia Computer Science, Vol. 220, 806-811.	"Optimization of decision making process in project management framework using hybrid method of particle swarm optimization and game theory" at The Twenty-Ninth International Symposium on Artificial Life and Robotics.
Main Technique	Particle Swarm Optimization (PSO)	Hybrid PSO + Imperialist Competitive Algorithm (ICA)	PSO + Game Theory for Auction Bidding	Metaheuristic algorithms for Feature Selection	Shapley Values for Explainability	Hybrid PSO + Game Theory for Decision-Making
Problem Addressed	Nonlinear optimization and Neural Network training	Resource-Constrained Project Scheduling (RCPSP)	Multi-round procurement optimization	Feature selection in high-dimensional datasets	Model Interpretability in ML Clustering	Project Management Decision-Making
Application Domain	Optimization Problems (General)	Construction Project Management	Auction Systems, Procurement	Feature Selection in ML	Machine Learning Clustering	Construction and Project Risk Management
Optimization Type	Single-objective Optimization	Multi-objective Scheduling Optimization	Multi-objective Optimization (Bidders and Costs)	Multi-objective Feature Optimization	Not Optimization; Explanation method	Multi-objective Decision Optimization
Metaheuristic / ML Method	Classical PSO (original)	Hybrid PSO-ICA	PSO enhanced with Game Theory (Nash Equilibrium)	Comparison of various metaheuristics (PSO, GA, ABC, FA, etc.)	Shapley Value Theory (Cooperative Game Theory)	PSO + Nash Equilibrium Principles
Innovation Level	First introduction of PSO concept	Hybridization for better RCPSP solving	Applying PSO to Game-Theoretic Procurement Problems	Extensive metaheuristic benchmarking for feature selection	New use of Shapley for Clustering Explainability	Hybrid PSO-Game Theory model tailored to construction
Strengths	Simple, flexible, computationally light	Better performance than single algorithms (PSO or ICA)	Provides win-win bidding solutions; balances fairness	Broad coverage; in-depth review; identifies key algorithm properties	Enhances explainability of unsupervised models	Considers strategic behaviors + optimization, improves project outcomes
Limitations	May get trapped in local optima	Hybrid models increase computational complexity	Computationally more expensive due to Nash search	No new algorithm proposed; only review	Not an optimization method	Still subject to local optima risk of PSO
Use of Game Theory	No	No	Directly (Nash Equilibrium)	No	Indirectly (Shapley Values)	Directly (Nash Equilibrium)
Feature Selection Focus	No	No	No	Yes	No	No
Explainability Focus	No	No	No	No	Yes (High)	No

Figure 2. Comparative Analysis of Studies Integrating Particle Swarm Optimization (PSO) and Large Language Models (LLMs): Technical Descriptions, Applications, and Challenges

Overall, the studies illustrate a clear trajectory: from foundational optimization methods to sophisticated hybrid systems and interpretability techniques. They reflect a broader trend in computational research — one that seeks not only to optimize performance but also to ensure fairness, transparency, and adaptability in increasingly dynamic environments.

3. Innovative Solutions: Introducing ECO InvestMind AI

3.1 Limitations of Existing Solutions and Our Proposed Approach

While existing AI-driven financial advisory platforms such as AlphaFin, FinRobot, and FinGPT have made significant contributions to the field of automated wealth management, they remain limited in their ability to provide fully personalized, comfortable, real-time, and behaviorally optimized financial planning solutions. Our proposed financial advisor represents a novel approach that bridges these gaps by integrating ML models with highly customized user experience, thereby offering a comprehensive and adaptive financial advisory platform tailored to individual users' evolving needs.

Investing in the financial markets involves significant risks, largely due to volatility and the influence of diverse external factors, including public sentiment driven by financial news. Traditional investment decision-making often overlooks or inadequately integrates the emotional and informational impact of news events, potentially leading to suboptimal financial decisions. Therefore, there is a need for advanced tools capable of accurately predicting stock prices by effectively incorporating both quantitative market data and qualitative sentiment analysis of news sources.

Although the AI-driven financial advisory systems exhibit significant advances in the application of ML to financial analysis, they fall short in several critical areas, including personalized financial planning, risk-adjusted portfolio recommendations, and real-time adaptive forecasting. Our proposed financial stock recommender, «ECO

InvestMind AI», aims to address these gaps by integrating cutting-edge ML algorithms with highly personalized user experience.

Key innovations of our proposed system include:

(1) advanced ML-driven investment diversification and nudging mechanisms, which is integrated in our platform integrates ML-based diversification strategies that optimize asset allocation in accordance with market conditions and user preferences; (3) real-time income and expense forecasting with visualization, which will help the financial advisor to generate predictive insights on future income and expenses, allowing users to make informed financial decisions; (4) cost optimization and recommender system for expense reduction, that identifies opportunities for cost savings, such as optimizing subscription services, selecting more cost-effective financial products, or adjusting lifestyle expenditures to align with savings goals.

Our objective is to develop a mobile stock recommendation system that leverages ML algorithms to provide precise stock price predictions. This application will uniquely integrate sentiment analysis derived from financial news, thus offering enhanced, context-aware investment recommendations. By combining quantitative stock market trends and qualitative news sentiment, the proposed application aims to deliver personalized, actionable insights to investors, improving decision-making quality and ultimately financial outcomes.

3.2 Software Structure and Functions of ECO InvestMind AI

The financial recommender application is structured according to the principles of Clean Architecture, which promotes modularity, maintainability, and scalability. It is organized into three primary layers that work together to ensure a robust and flexible software system. Its software application can be seen in Figure 3.

At the forefront is the Presentation Layer, which is developed using Android's Jetpack Compose. This layer is responsible for managing user interactions through an intuitive interface and is connected to the ViewModel. The ViewModel facilitates the presentation of real-time updates using LiveData, ensuring seamless and responsive user experiences.

The Domain Layer contains the core business logic of the application. It is centered around well-defined use cases that encapsulate key functionalities, such as generating stock recommendations via the StockRecoUseCase and listing available stocks through the StockListUseCase. This separation of concerns enhances code clarity and testability, supporting the application's long-term maintainability.

Finally, the Data Layer manages data acquisition, processing, and persistence. It includes an API network component designed to fetch real-time financial information from external sources. Additionally, local storage mechanisms enable data caching and offline access, ensuring that user data remains synchronized even when the application is used without an internet connection. The repository implementation serves as an intermediary between the data sources and the domain use cases, establishing a clean and consistent data flow across the application.

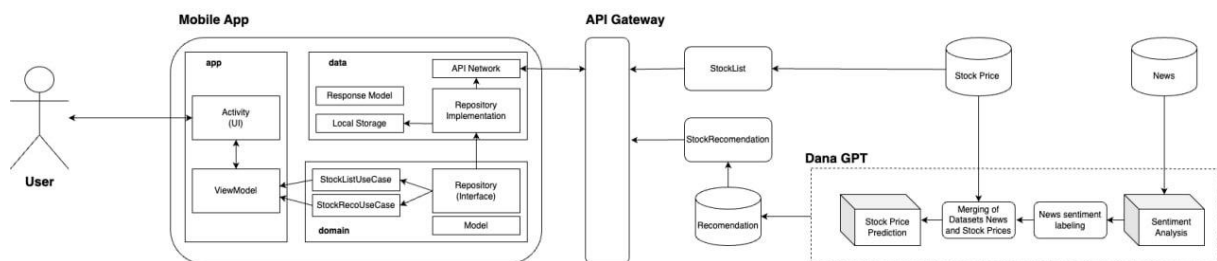


Figure 3. ECO InvestMind AI Software Architecture

Our solution employs a layered enterprise architecture composed of three primary layers: the Presentation Layer, Business Logic Layer, and Data Layer (Figure A1 from Appendix A).

The Presentation Layer, situated at the top of the architecture, interacts directly with the users through a user-friendly interface designed using Android Jetpack Compose. This layer comprises two main components. The User Activity component provides an intuitive graphical user interface, enabling users to access key application features effortlessly. Additionally, the View Model acts as an intermediary, processing and transforming data received from the Business Logic Layer into observable states, thus facilitating reactive and real-time user interaction.

The Business Logic Layer houses the core functionality and rules of the Robo-advisor. It includes the Stock-RecoUseCase, responsible for generating personalized stock recommendations based on predictive models, and the StockListUseCase, which manages operations related to fetching and presenting lists of available stocks to users. This layer also features a Repository Interface, which abstracts data access, allowing seamless retrieval without dependency on specific data sources, and the Model component, which applies ML algorithms to provide accurate stock price predictions and sentiment-based insights.

The Data Layer manages and prepares data, ensuring its readiness for processing and analysis. It consists of Database Tables storing historical stock price data, ESG scores, and news sentiment analysis results. Additionally, Store Process/Views facilitate essential database operations and maintain updated views for efficient querying. The Data Preprocessing component is responsible for cleaning, transforming, and preparing raw data sourced from CSV files for ML tasks. Finally, Data CSV Files contain initial datasets such as historical stock prices, ESG scores, and news sentiment data utilized in training and analysis.

This layered architectural approach ensures clear separation of concerns, enhances system modularity, facilitates easier maintenance and testing, and provides the scalability necessary to meet evolving user demands and increasing data complexities.

A recommendation system for investing in stocks collects historical stock price data, news, and ESG information from large companies represented on the Kazakh and American stock markets. The data comes from web portals kapital.kz, kase.kz, finance.yahoo.com, as well as from the official websites of the companies themselves to obtain basic information such as financial statements, company structure and strategic goals.

Additionally, ESG scores and ratings are downloaded from specialized platforms such as lseg.com. Analyzing the data obtained using AI and ML methods, the program predicts stock price changes for the future 7 days and generates personalized recommendations for the user to buy or sell shares.

The Bot at the bottom of the diagram updates the data in the datasets every week as new data appears on the news websites and new stock prices are collected on the appropriate site.

3.3 Mobile Application Architecture of Our System

An application architecture of our mobile application based on Kotlin Multiplatform (KMP) is shown at Figure 4. Our goal is to create a single, reusable codebase that works efficiently across both Android and iOS platforms, reducing code duplication while maintaining native user experiences.

The core of the application is built upon a shared Kotlin module, which serves as the foundation for both Android and iOS platforms. This shared module includes clearly defined use cases that encapsulate the core application logic and manage user interactions. A unified repository layer is implemented to handle data from both remote and local sources, utilizing the Ktor library for seamless network communication and SQLite for reliable offline data storage. This architecture promotes consistency and reusability across platforms.

To ensure predictable and consistent user interface behavior, the application adopts the Model-View-Intent (MVI) architectural pattern for UI state management. This approach supports unidirectional data flow, which enhances the maintainability and testability of the UI logic.

For platform-specific features such as GPS, camera access, and sensor data, the application leverages Kotlin's expect/actual mechanism. This interoperation layer allows clean and efficient integration of the shared Kotlin logic

with native Android and iOS APIs, maintaining platform fidelity while maximizing code reuse.

On the native side, the Android interface is developed using Jetpack Compose, which enables the construction of modern and responsive user interfaces. For iOS, SwiftUI is utilized, ensuring full integration with native iOS SDK components. When a user interacts with the application, the platform-specific UI communicates with the shared module via the interop layer. The resulting data and logic flow back to the native UI, ensuring a smooth and responsive user experience. This design approach enhances development efficiency, simplifies maintenance, and leverages the unique capabilities of each platform.

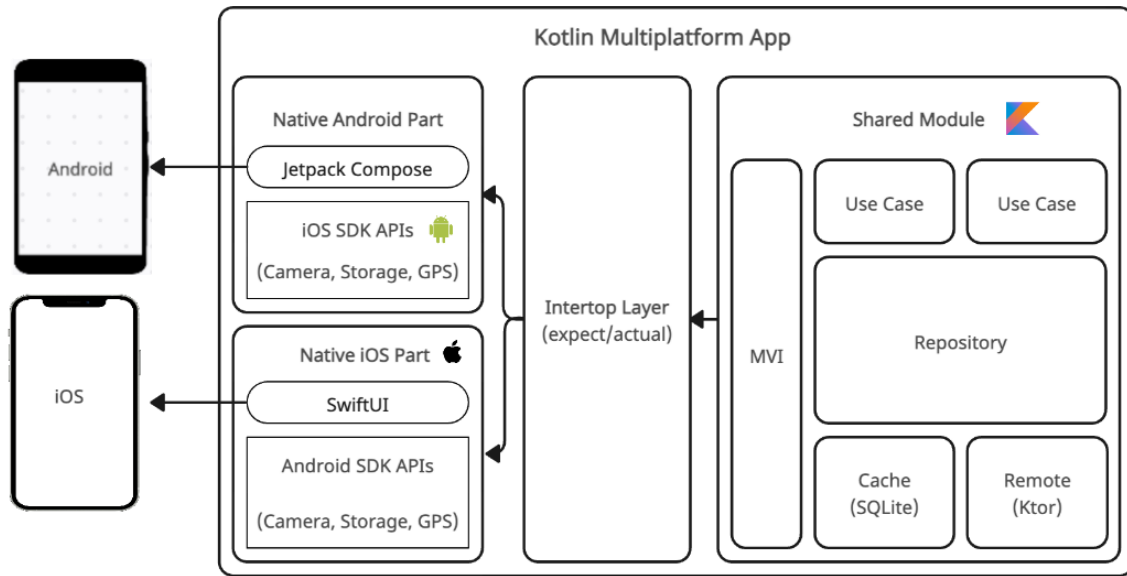


Figure 4. Kotlin Multiplatform Application Architecture

The primary scope of the application lies in providing a stock investment recommendation system designed for use in financial markets. It delivers detailed company information and offers buy or sell recommendations based on stock price forecasts generated through ML algorithms. The system automatically collects the latest financial news by parsing content from web portals such as Kapital.kz and Finance.Yahoo.com. It employs a BERT-based neural network, utilizing transformer architecture for sentiment analysis to determine the emotional tone of each article—categorizing content as positive, neutral, or negative.

Based on the sentiment analysis results, the application utilizes a regression-based neural network to forecast short-term stock prices. These forecasts are then used to generate informed investment recommendations, enabling users to make better financial decisions. The methodology integrates a structured data pipeline, advanced ML algorithms, and a mobile application framework to deliver a personalized financial advisor platform. This approach incorporates database design, data processing workflows, and software architecture principles to ensure robustness, scalability, and user-centric functionality.

3.4 Database Design and Datasets Description of ECO InvestMind AI

The database, shown at Figure A2 in Appendix A, designed to support the financial recommender application

integrates heterogeneous data sources to enable comprehensive investment decision-making. The schema is organized around three primary data domains—stock market prices, financial news sentiment, and Environmental, Social, and Governance (ESG) metrics—complemented by static company attributes and predictive model outputs. This relational database structure facilitates complex analytical workflows by linking time-series financial data with textual and sustainability indicators.

Stock price data is stored in a series of normalized tables representing different markets and individual tickers, such as KASE, NASDAQ, S&P 500, and specific company stocks including KSPI, HSBK and KZAP. Each stock price table contains daily records with attributes such as open, close, high, low, and volume. The combination of Ticker and Date serves as the composite primary key, allowing precise temporal alignment and historical tracking across exchanges. These price datasets were sourced from finance.yahoo.com, covering the period from 2020 to 2025.

Each stock price table is linked to a corresponding news table through foreign key relationships on the Ticker and Date fields. These news tables contain textual content related to the associated company or market, the date of publication, and sentiment labels assigned through automated sentiment analysis algorithms. The sentiment label is stored as an integer value, reflecting whether the tone of the news is negative, neutral, or positive. This alignment between news sentiment and trading data enables joint analysis of market movements in response to narrative signals, enhancing predictive capabilities.

In addition to market and sentiment data, the schema includes a dedicated ESG table that aggregates sustainability-related metrics for each company, indexed by the Ticker field. The table captures a wide range of environmental, social, and governance indicators including emissions, innovation, product responsibility, human rights, shareholder engagement, and overall ESG scores. These quantitative metrics provide a foundation for evaluating firms based on ethical and long-term risk criteria. The ESG table is structurally connected to the stock price and company information tables, creating a unified profile for each listed entity that spans financial and non-financial dimensions.

Static firm-level attributes are stored in the Company Basic Information table, which includes fields such as company name, market capitalization, industry classification, foundation year, dividend income, employee count, and headquarters location. This table shares the Ticker key with other core tables, providing a metadata layer for contextualizing financial performance and sustainability assessments.

Forecasting results generated by the system's ML models are captured in the Results table. Each record contains a predicted close price, prediction accuracy score, confidence intervals (upper and lower bounds), forecasted price trend, cost at risk, maximum drawdown, volatility indicators, and general risk assessments. These prediction outputs are tied to specific tickers and dates, aligning them with historical prices and sentiment events for validation and application. The data feeding these models is drawn from the integrated ecosystem of price, news, and ESG information, enabling multivariate time-series prediction and scenario analysis.

The relational model ensures referential integrity through consistent use of foreign key constraints, establishing direct links between time-indexed financial data and corporate-level features. News entries are aligned with corresponding stock price entries via shared date and ticker identifiers. ESG and company information are mapped to price and prediction records through ticker-based joins, allowing longitudinal and cross-sectional analyses of firm behavior. This architecture supports advanced analytics and real-time recommendation services by facilitating unified access to historical patterns, market sentiment, sustainability evaluations, and forecast data within a single, scalable data infrastructure.

The datasets used in *ECO InvestMind AI* comprise a diverse set of structured and unstructured data sources that enable comprehensive financial analysis and predictive modeling. The detailed description of dataset is shown in Figure 5.

Dataset	Number of Rows	Number of Columns	Main Features	Data Type
ESG	14	16	Company-level ESG indicators across Environment, Social, and Governance categories. Includes scores for emissions, resource use, human rights, CSR strategy, and a composite ESG score.	Structured numerical and categorical data
Company Basic Information	17	8	Includes static firm-level attributes such as market capitalization, industry, foundation year, dividend income, employee number, and headquarters. Provides a metadata layer for contextualizing financial performance and ESG alignment.	Structured categorical and numerical firm-level data
Results	7	11	Contains predictive outputs from ML models including forecasted close price, prediction accuracy, confidence intervals, risk metrics, and investment category labels. Used for investment recommendations and portfolio analysis.	Structured numerical and categorical output from predictive models
News (for every company)	1270	4	Historical news data for company (Date, Ticker of company, News text, Sentiment label)	Textual news data with categorical sentiment label
Stock Prices (for every company)	1270	7	Historical stock prices data for company including trading metrics (open, close, volume, etc.)	Numerical data with datetime and textual column

Figure 5. Summary of Datasets Used in ECO InvestMind AI System

The ESG dataset (14×16) provides firm-level sustainability indicators across environmental, social, and governance dimensions. Company Basic Information (17×8) includes static metadata such as industry, market capitalization, and foundation year. The Results dataset (7×11) contains model-generated forecasts with associated accuracy metrics, risk indicators, and recommendation labels. Additionally, the News dataset (1270×4) offers timestamped financial news with sentiment labels, while the Stock Prices dataset (1270×7) presents historical trading data for each company. Together, these datasets support sentiment analysis, ESG integration, and stock price prediction for investment decision-making.

3.5 Layered Architecture of ECO InvestMind AI Backend System

The backend system of ECO InvestMind AI is developed using the NestJS framework, structured around a layered modular architecture to ensure separation of concerns, reusability, and maintainability. This architecture is divided into four distinct layers: the Module Layer, Controller Layer, Service Layer, and Storage Layer, each contributing to the logical and operational flow of the system. The detailed backend structure is shown in Figure 6.

At Module Level the App Module serves as the entry point and coordination hub for the backend. It integrates and configures all feature-specific modules, namely:

- Auth Module
- User Module
- Yahoo Finance Module

This layer also establishes the application's connection to the MongoDB database using Mongoose, and loads global configurations. Each feature module encapsulates its domain logic and dependencies, promoting a loosely coupled system design that adheres to the principles of domain-driven development.

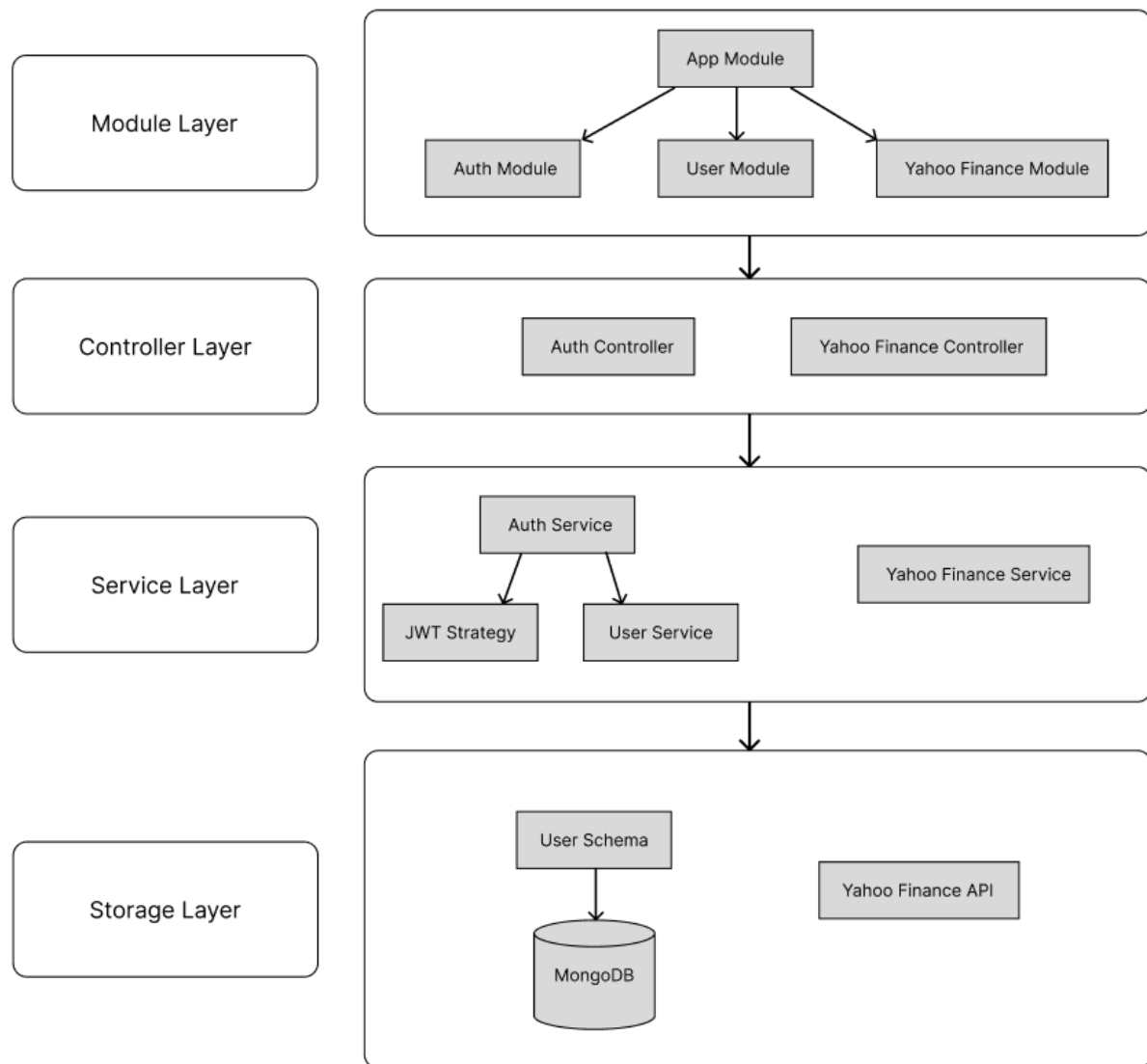


Figure 6. Backend Structure

The controller layer exposes the public-facing HTTP interfaces and defines the system's routing structure. It includes:

- Auth Controller, which handles endpoints for user registration, authentication, and secure access token management;
- Yahoo Finance Controller, which facilitates the retrieval of external financial data such as stock prices and index metrics.

Controllers are responsible for handling incoming requests, validating inputs, and forwarding business logic execution to the underlying service layer.

Service Layer is dedicated to encapsulating the application's core business logic and orchestrating domain-specific workflows. Key services include:

- Auth Service, which manages user authentication and token-based access control. It integrates:
- A JWT Strategy for secure token validation;

- The User Service, which offers user-related data access and utility operations.
- Yahoo Finance Service, which communicates with the external Yahoo Finance API to retrieve, process, and transform financial data into application-ready formats.

These services are designed for high cohesion and testability, with clear separation from infrastructure concerns.

The storage layer handles persistence and external data integration:

- The User Schema defines the MongoDB schema for storing user credentials and metadata;
- The MongoDB database serves as the primary data store;
- The Yahoo Finance API is leveraged as an external data provider to support the application's financial analytics functionality.

This layer is abstracted from the business logic through schema-driven modeling and repository patterns, ensuring robustness and data consistency.

The modular architecture of ECO InvestMind AI aligns with best practices in software engineering by facilitating modular growth, ease of testing, and scalability. Modules are decoupled yet interoperable via explicit contracts and shared services. This enables future extensions—such as new financial data providers or additional authentication mechanisms—without requiring substantial modifications to existing modules.

By leveraging the layered structure and NestJS's dependency injection system, the architecture supports clean code organization, enhanced maintainability, and improved separation of responsibilities across the system.

3.6 Data Privacy, Use, Sharing, and Regulatory Compliance

ECO InvestMind AI collects only the minimum data needed to power personalized recommendations and forecasts. Public market information—historical stock prices, ESG scores, and news headlines—is drawn from open sources under their usual terms of service (Yahoo Finance, n.d.; LSEG, n.d.; Kapital.kz, 2024). From each user, we only gather an email address (as a unique ID), basic profile details, and investment parameters (initial capital, monthly contribution, risk tolerance). No bank-card or other payment credentials are ever stored.

All inputs flow exclusively through our recommendation engine, sentiment-analysis and forecasting models, and into a secured backend governed by strict role-based access controls. Outside the system, we publish only fully anonymized, aggregate metrics—such as overall model accuracy. Data in transit are protected by HTTPS/TLS 1.2 (Rescorla, 2008) and data at rest are encrypted with AES-256 (NIST, 2001). On Android we use Jetpack Encrypted DataStore with certificate pinning (Android Developer, n.d.); on iOS secrets reside in the Keychain or Secure Enclave under App Transport Security (Apple Developer, n.d.). All production-database access is logged and audited.

User data remain only while an account is active and are purged within 30 days of deletion (Republic of Kazakhstan, 2013). We comply with GDPR (European Parliament, 2016) and Kazakhstan's Law on Personal Data (Republic of Kazakhstan, 2013); users may request access, correction, or erasure via in-app settings or support.

ECO InvestMind AI also follows Kazakhstan's regulations on investment services and financial advice, notably the Securities Market Law (Republic of Kazakhstan, 2007) and National Bank directives. Integrations with brokerage platforms use only official APIs and meet all licensing and transaction-record-keeping requirements to ensure transparency. The app recommends brokers based on their licenses, fees, and features, but all actual trades—buying, selling, account opening—are executed directly through the brokers' own interfaces; ECO InvestMind AI serves purely as an informational assistant.

4. ML-Engine Behind: AI Assistant based on Advanced ML models

4.1 Hybrid ML Architecture Overview

The core of the financial advisor system is a hybrid ML architecture named Dana, specifically designed to provide accurate and context-aware stock price forecasts for companies. The architecture integrates time series forecasting and natural language processing (NLP) techniques to incorporate both quantitative market data and qualitative financial sentiment into its predictive modeling.

The model architecture is organized into a dual-stream pipeline. The first stream focuses on sentiment analysis. Financial news headlines are collected, preprocessed, and passed through a fine-tuned BERT-based transformer model that performs sentiment classification. Each news item is assigned a sentiment label—positive, neutral, or negative—reflecting the emotional tone of the content. These sentiment labels are then aligned temporally with corresponding financial series data to ensure consistency in downstream analysis.

The second stream involves time series forecasting based on historical market data. Specifically, the system utilizes OHLCV (Open, High, Low, Close, Volume) features from KASE-listed stocks and feeds them into a MLP neural network. MLP is selected due to its proven ability to capture sequential dependencies and model market trends, volatility, and fluctuations over time.

The outputs of the sentiment analysis and MLP forecasting streams are subsequently integrated to produce the final predictive output. This fusion approach allows the model to dynamically adjust its forecasts in response to sentiment shifts observed in financial news, particularly during periods of increased market sensitivity to external events. By combining structured price data with unstructured textual sentiment, the Dana architecture improves forecast accuracy and robustness in the presence of news-driven volatility. In addition to generating predicted stock prices, the system also computes supplementary metrics such as confidence intervals, risk assessments, and volatility indicators to support more informed investment decisions.

Moreover, the application offers users an additional service of building the stock portfolio. Users will input investment amount, initial capital and desired risk score. Then the model will construct a stock portfolio tailored to user-defined investment parameters using historical return data and Particle Swarm Optimization (PSO). The system is designed to help individual investors achieve long-term financial goals by recommending an optimal asset allocation strategy based on return maximization.

4.2 ML Model Selection and Hyperparameters

In the design of the proposed system, careful consideration was given to the selection of ML models that could effectively capture both qualitative and quantitative aspects of financial decision-making. To this end, two complementary models were adopted: BERT for financial news sentiment analysis and a Multilayer Perceptron (MLP) for stock price forecasting. This selection was informed by a review of recent literature, which highlights the superior performance of these models in comparable financial and predictive contexts.

BERT (Bidirectional Encoder Representations from Transformers), introduced by Devlin et al. (2019), has established itself as a benchmark in natural language processing due to its ability to model deep bidirectional representations of text. Its contextualized word embeddings allow for the nuanced understanding of sentiment and intention, which are critical when analyzing financial news where subtle shifts in tone can drive market behavior. Numerous recent studies (Sayeed, Mohan, & Muthu, 2023; Wu, Jin, Shi, Liang, & Zhan, 2024) have demonstrated BERT's effectiveness in sentiment analysis tasks, consistently outperforming earlier approaches. Furthermore, Ren (2024) showed that incorporating multimodal data sources alongside BERT further improves sentiment prediction accuracy. Domain-specific enhancements, such as Opinion-BERT (Hossain, Hossain, Mridha, Safran, & Alfarhood, 2025) and BERTCaps (Memari, Nejad, Rabiei, Eisaei, & Hesarakhi, 2024), highlight the model's adaptability to specialized applications. Given its demonstrated performance and robustness, BERT was selected as the core model for processing unstructured textual data within this research.

In parallel, for the structured task of stock price forecasting, a Multilayer Perceptron (MLP) was selected. First formalized by Rumelhart, Hinton, and Williams (1986), MLPs are a class of feedforward neural networks capable of capturing complex nonlinear relationships, an essential requirement in the inherently noisy and volatile environment of financial markets. Zhang, Eddy Patuwo, and Hu (1998) emphasized the strength of MLPs in approximating any continuous function, which makes them particularly well-suited for modeling market behaviors that are often non-linear and dynamic. Recent studies have further validated this choice: Abotaleb et al. (2024) showed that MLP architectures excel in stock price classification under conditions of economic uncertainty, while Gruver et al. (2024) demonstrated their contribution to multi-stage stock forecasting pipelines. Moreover, Swenson (2024) applied neural network models to successfully forecast corporate quarterly tax payments, confirming MLP's applicability to financial time series data. The effectiveness of MLPs in earnings prediction was further reinforced by Kuryłek (2024), who applied MLP-based models to earnings per share (EPS) forecasting on the Warsaw Stock Exchange, and by Xie, Xu, Yan, Qian, and Yang (2024) in the context of multi-country GDP prediction tasks.

By combining BERT for sentiment extraction and MLP for numerical trend forecasting, the proposed system adopts a hybrid dual-stream architecture. This design enables the ECO InvestMind AI application to synthesize insights from both real-time qualitative market sentiment and quantitative price dynamics. Such an integrated approach aims to enhance real-time adaptability, improve personalization of investment strategies, and ultimately support more informed financial decision-making in dynamic market environments.

The pseudocode of the system can be found in Figure 7.

All experiments were conducted using Python 3.10 with the following libraries and frameworks:

- Scikit-learn (v1.3) — for preprocessing, MLP modeling, and evaluation.
- Transformers (Hugging Face) (v4.39) — for BERT tokenizer and BERT fine-tuning.
- TensorFlow/Keras — for additional deep learning model support.
- PyTorch — used indirectly through the Hugging Face Trainer API for BERT.
- Pandas, NumPy, Matplotlib, Seaborn — for data manipulation and visualization.
- Computation was carried out on a standard Google Colab Pro environment, providing Tesla T4 GPU acceleration for BERT training, and CPU runtime for MLP regression.

The selected models were configured with the following hyperparameters:

BERT for Sentiment Classification:

- Model: bert-base-uncased
- Maximum input length: 128 tokens
- Training epochs: 3
- Batch size: 16 (train and evaluation)
- Optimizer: AdamW (default in Trainer API)
- Learning rate: 5e-5 (default)
- Evaluation metric: Accuracy
- Save Strategy: Save the best model by accuracy after each epoch.

MLPRegressor for Stock Price Forecasting:

- Architecture: 2 hidden layers with (32, 16) neurons respectively.
- Activation function: ReLU
- Maximum iterations: 2000
- Random state: 42 (for reproducibility)
- Optimizer: LBFGS (default solver in MLPRegressor for small datasets).

- Normalization: StandardScaler was applied to features before training.

Algorithm 1 AI-Powered Wealth Forecasting and Portfolio Planning

```

1: Input: News dataset, historical stock data
2: Output: Forecasted prices, investment advice, stock allocation

3: // BERT-Based Sentiment Classification
4: Map sentiment labels:  $\{-1, 0, 1\} \rightarrow \{0, 1, 2\}$ 
5: Split data into 80% train / 20% test
6: Load bert-base-uncased with 3-class output layer
7: Fine-tune with:
8:   Epochs = 3, Batch size = 16, Learning rate = default AdamW
9: Evaluate using accuracy, precision, recall, F1-score
10: Predict sentiment label for new financial news headline

11: // Stock Price Forecasting with Neural Network
12: Extract date features: Day, Month, Weekday
13: Use StandardScaler to normalize input features
14: Train MLPRegressor with:
15:   Hidden layers = (32, 16), Activation = ReLU, Max Iter = 2000
16: Forecast closing prices for dates: April 1-7, 2025
17: Compare with true prices using MAE, RMSE, Accuracy

18: // Forecast Evaluation and Risk Profiling
19: Compute:
20:   Trend direction (Upward / Downward)
21:   95% Confidence Interval:  $\mu \pm 1.96\sigma$ 
22:   Volatility = std deviation of predicted prices
23:   Cost-at-risk and Maximum drawdown
24: Classify risk: Low, Moderate, High
25: Assign category: Conservative / Balanced / Aggressive

26: // Portfolio Optimization with PSO
27: Calculate annualized return for each asset
28: Initialize swarm with:
29:   Particles = 30, Iterations = 100, Dimensions = #stocks
30:   Inertia = 0.5, Cognitive = 1.5, Social = 1.5
31: for each iteration do
32:   for each particle do
33:     Update velocity and position based on best scores
34:     Objective: minimize  $|\text{TargetGoal} - \text{FutureValue}|$ 
35:   end for
36: end for
37: Normalize weights  $\rightarrow$  final asset allocation

38: // Simulate Monthly Investment Plan
39: for each stock in allocation do
40:   Compute budget share = monthly_investment  $\times$  weight
41:   Calculate quantity of shares to buy based on latest price
42: end for
43: Compute 36-month portfolio quantities1

44: // Market Sentiment Index Calculation
45: for each stock do
46:   Compute:
47:     Price change %, Volume change %, Intraday volatility
48:     Normalize all indicators to [0, 100]
49:     Invert volatility score (low volatility = positive sentiment)
50: end for
51: Sentiment Index = average of normalized values
52: Output final market mood score (0 to 100)

```

Figure 7. Pseudocode of ECO InvestMind AI

The choice of these hyperparameters was guided by a combination of best practices found in literature and

empirical performance optimization during initial experimentation phases.

4.3 ML Models Description and Mathematical Foundations

4.3.1 News Sentiment Classification using BERT model

To analyze and predict stock movement, the sentimental analysis of the news about the companies will be done by employing a two-stage ML framework combining NLP with regression modeling. The first component of the system is a sentiment classification model based on BERT.

This model is fine-tuned to classify financial news into three sentiment categories: positive, neutral, and negative. Let T_i represent a financial news headline. The BERT architecture converts T_i into a sequence of contextual embeddings, which are processed through multiple transformer layers to produce a probability distribution over sentiment classes. Mathematically, this can be expressed as:

$$\hat{y}_i = \text{softmax}(f_{BERT}(T_i)) \in \mathbb{R}^3 \quad (1)$$

where f_{BERT} is the output layer of the BERT model, and $\hat{y}_i \in \mathbb{R}^3$ denotes the predicted probabilities for each class. The predicted label corresponds to the class with the highest probability, $\text{argmax}_j \hat{y}_{i,j}$.

The model is trained to minimize the cross-entropy loss between the predicted and actual labels,

$$L_{BERT} = - \sum_{i=1}^N \sum_{j=1}^3 y_{i,j} \log(\hat{y}_{i,j}), \quad (2)$$

where $\hat{y}_{i,j}$ is the true label for instance i .

4.3.2 Stock Price Forecasting using MLP Regressor

The second component of the system is a MLP Regressor, which is designed to predict stock closing prices. The input to the MLP consists of numerical features (such as open, high, low prices and volume) as well as the sentiment label predicted by BERT.

Let $x_i \in \mathbb{R}^d$ be the feature vector for instance i , and $y_i \in \mathbb{R}^d$ be the target closing price. The MLP is a feedforward neural network that applies a sequence of linear transformations and nonlinear activations.

The hidden layers are defined as

$$\begin{aligned} h^{(1)} &= \text{ReLU}(W^{(1)}x_i + b^{(1)}), \\ h^{(2)} &= \text{ReLU}(W^{(2)}h^{(1)} + b^{(2)}), \\ \hat{y}_i &= W^{(3)}h^{(2)} + b^{(3)}, \end{aligned} \quad (3)$$

where ReLU is the ReLU activation function, $W^{(l)}, b^{(l)}$ are learnable weights and biases for layer l , $\hat{y}_i$ is the predicted stock closing price.

The training objective is to minimize the Mean Squared Error (MSE) between the predicted and actual closing prices:

$$L_{MLP} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (4)$$

Model performance is further evaluated using metrics such as Root Mean Squared Error (RMSE), defined as $\sqrt{MSE}$, and Mean Absolute Error (MAE), given by:

$$\frac{1}{N} \sum |y_i - \hat{y}_i| \quad (5)$$

This hybrid architecture leverages both textual and numerical signals to deliver more context-aware and accurate financial predictions.

4.3.3 Stock Portfolio Builder with PSO

Additionally, stock portfolio building is complete pipeline using historical return data and optimization through PSO. It is designed for individual investors seeking to grow their wealth by making informed, data-driven allocation decisions based on past performance.

To estimate the annualized return for each asset, the code employs the compound annual growth rate formula. For each stock i , the annual return is computed as:

$$R_i = \left(\frac{P_{end,i}}{P_{start,i}} \right)^{\frac{1}{T_i}} - 1, \quad (6)$$

where $P_{end,i}$ and $P_{start,i}$ are the stock's prices at the start and end of the dataset, and T_i represents the time period in years between the two prices, calculated from the number of calendar days. This process yields a dictionary of expected annual returns that serves as the core input for the subsequent optimization step.

The user is assumed to provide an initial capital investment and a monthly contribution over time period in months. The objective is to determine the optimal portfolio weights $w = [w_1, w_2, \dots, w_n]$, that maximize the total capital at the end of the investment period.

The expected annual portfolio return is computed as:

$$R_{portfolio} = \sum_{i=1}^n w_i * R_i, \quad (7)$$

Assuming continuous compounding with reinvested monthly contributions, the future value of the investor's portfolio after t months is approximate as:

$$V(t) = C_0 * (1 + R_{portfolio})^{t/12} + I * t * (1 + R_{portfolio})^{t/12}, \quad (8)$$

where C_0 is the initial capital and $I * t$ is the total of monthly contributions by month t . The objective function used in the PSO algorithm is therefore:

$$f(w) = -V(t) = - \left[C_0 * (1 + R_{portfolio})^{T/12} + I * t * (1 + R_{portfolio})^{T/12} \right], \quad (9)$$

where T = selected time period. Since the PSO algorithm minimizes the objective function, the negative sign ensures that the maximum future value is achieved by minimizing its negative.

The optimization algorithm begins with a population of $N = 30$ particles, each representing a potential

allocation vector $w_j \in \mathbb{R}^n$. Each particle has a velocity vector $v_j \in \mathbb{R}^n$ and is updated iteratively using the standard PSO update rules:

$$\begin{aligned} v_j &\leftarrow w * v_j + c_1 * r_1 * (p_j - w_j) + c_2 * r_2 * (g - w_j), \\ w_j &\leftarrow w_j + v_j, \end{aligned} \quad (10)$$

where:

- p_j is the personal best position for particle j ,
- g is the global best across all particles,
- w , c_1 and c_2 are inertia and acceleration constants,
- r_1 and r_2 are randomly sampled.

After each particle's position is updated during optimization, the values are adjusted to stay within valid limits and ensure that the portfolio weights add up to 100%. This process is repeated for 100 iterations. At the end, the best-performing set of weights found during the process is chosen as the final recommended investment allocation.

Using these final portfolio weights, the system then simulates how the investor's capital would grow over a given time period (e.g. 36 months). It calculates the total wealth for each month, taking into account both the initial capital and the monthly contributions. These results are stored in a time series and visualized in a graph to provide a clear view of how the investment is expected to grow over time.

The output consists of a table showing the estimated annual return for each stock and how much of the portfolio should be invested in each one. The capital growth graph is also labeled with milestone values every six months, helping the investor see the progression of their wealth.

4.3.4 Market Sentiment Evaluation

To evaluate the prevailing market mood, a Market Sentiment Index was developed by integrating three key indicators derived from selected stocks: the price change between two consecutive days, the volume change between two consecutive days, and the volatility, which was measured as the relative difference between the daily high and low prices in relation to the opening price.

Each of these indicators was normalized to a common scale ranging from 0 to 100, ensuring comparability across different measures. In the case of price and volume changes, a higher normalized value was interpreted as indicative of a more positive market sentiment. Conversely, for volatility, the normalized score was inverted so that lower volatility would contribute positively to the overall sentiment evaluation.

The normalization procedure employed the standard min-max scaling method, as expressed by the following formula:

$$Normalized\ Value = \frac{Value - Minimum}{Maximum - Minimum} * 100 \quad (11)$$

For volatility, the adjustment was made according to the formula:

$$Volatility\ Index = 100 - Normalized\ Volatility \quad (12)$$

After normalization, the three components—the Price Index, Volume Index, and Volatility Index—were

averaged to obtain a single Sentiment Index for each individual stock. Subsequently, the Market Sentiment Index for the selected date was determined by calculating the average sentiment value across all stocks, following the expression:

$$\text{Market Sentiment Index} = \frac{1}{n} \sum_{i=1}^n \text{Sentiment Index}_i, \quad (13)$$

where n represents the number of stocks analyzed.

4.4 Results: News Sentiment Analysis and Stock Prices Prediction

The experimental evaluation of the hybrid ML pipeline was conducted on five major securities listed on the Kazakhstan Stock Exchange: KASE index, HSBK (Halyk Bank), KSPI (Kaspi.kz), KEGC (KEGOC), and KZAP (Kazatomprom). These companies represent a cross-section of financial, industrial, and utility sectors and were selected to evaluate model generalizability across domains. The models were tested on historical data spanning from January 2020 to March 2025, with forecasts conducted for the first week of April 2025.

Company/ Index	BERT Model				MLP Model			
	Accuracy	F1	Precision	Recall	MSE	RMSE	MAE	Accuracy
KASE	99,07%	0,9906	0,9908	0,9907	8079,75	89,88	85,17	98,48%
HSBK	96,85%	0,9628	0,9572	0,9685	1014,04	31,84	26,57	91,09%
KSPI	98,89%	0,9890	0,9892	0,9890	8697423,27	2949,13	2753,29	94,05%
KEGC	98,82%	0,9823	0,9765	0,9882	22,56	4,75	3,87	99,74%
KZAP	99,60%	0,9960	0,9961	0,9960	2001163,93	1414,62	1386,66	92,03%

Table 1. ML Models results on News Sentimental Analysis and Stock Prediction

The BERT-based sentiment classifier consistently achieved high levels of accuracy and F1-score across all evaluated companies. As shown in Table 1, classification accuracy ranged from 96,85% (HSBK) to 99,60% (KZAP), with F1-scores between 0,9628 and 0,9960. Precision and recall remained similarly strong, with minimal deviation between classes. These results demonstrate the model's capacity to extract sentiment reliably from news headlines—a crucial capability given the influence of investor psychology and media narratives on stock market dynamics. The consistently high precision and recall values reinforce that the model is not only accurate but also balanced in its prediction quality across different sentiment categories.

These findings support the first research objective: integrating real-time news sentiment into predictive modeling to better reflect behavioral market shifts. The model's performance confirms that even relatively short financial news snippets can yield useful emotional indicators when processed through advanced natural language understanding models.

The MLP model exhibited solid performance in forecasting short-term stock price movements. Accuracy, measured as the directionally correct prediction of stock price movement, was above 91% for all companies and peaked at 99,74% for KEGC. However, performance varied significantly in terms of error metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE).

The lowest prediction error was achieved for KEGC, with an MSE of 22,56, RMSE of 4,75, and MAE of 3,87, suggesting highly stable pricing patterns and effective learning from historical data. Conversely, KSPI and KZAP presented higher volatility and greater prediction difficulty, as evidenced by large RMSE values (2949,13 and 1414,62, respectively). These results are consistent with real-world market characteristics—Kaspi.kz, for example, often experiences large price swings due to fintech sector speculation and media coverage.

Despite error variation, the model successfully captured trend direction in 4 out of 5 stocks and provided meaningful volatility and risk estimations, which were later used to classify investment profiles as conservative, balanced, or aggressive. To evaluate the overall performance of the hybrid system, we calculated the average accuracy of both the BERT-based sentiment classification model and the MLP-based stock price forecasting model across the five selected companies. The BERT model achieved high classification accuracy on each dataset: 99,07% (KASE), 96,85% (HSBK), 98,89% (KSPI), 98,82% (KEGC), and 99,60% (KZAP). The average classification accuracy of BERT was computed as:

$$BERT\ Accuracy_{avg} = \frac{99,07 + 96,85 + 98,89 + 98,82 + 99,60}{5} = 98,65\% \quad (14)$$

Similarly, the MLP model yielded directional prediction accuracies of 98,48% (KASE), 91,09% (HSBK), 94,05% (KSPI), 99,74% (KEGC), and 92,03% (KZAP). The average forecasting accuracy was calculated as:

$$MLP\ Accuracy_{avg} = \frac{98,48 + 91,09 + 94,05 + 99,74 + 92,03}{5} = 95,08\% \quad (15)$$

To estimate the overall performance of the proposed system, we define Total System Accuracy as the mean of the average BERT and MLP accuracies:

$$Total\ System\ Accuracy = \frac{98,65 + 95,08}{2} = 96,87\% \quad (16)$$

This result confirms that the dual-model architecture is highly effective, achieving a combined system-level accuracy of 96.87%, which reflects the strength of combining both sentiment and time-series signals in financial prediction.

4.5 AI Model Server Deployment and Hardware Configuration

To serve the AI model, a dedicated server is used. The server runs a Docker container with a FastAPI-based Python application. This application loads the AI model from a pre-trained .pt file and provides an API endpoint for interaction.

A second Docker container runs the NestJS backend, which sends inference requests to the Python server. This two-container setup helps separate concerns and improves system stability.

The performance of the AI model depends on both CPU and GPU. The CPU handles running the containers and managing network requests, while the GPU speeds up the model inference. A stronger GPU reduces response time and increases performance. If a GPU is not available, the model runs on the CPU, but the execution becomes much slower.

The server used for AI model deployment has the following specifications: GPU: NVIDIA RTX 3090 (24 GB)

- CPU: AMD Ryzen 5 5500
- RAM: 16 GB DDR4
- Operating System: Linux Ubuntu

4.6 User Experience: App Interface and Features

The developed mobile application provides a user-centric interface that simplifies the process of stock market monitoring and personal investment management. As shown in Figures 8 and 9, the application is designed to support both novice and experienced investors through intuitive navigation and visually informative components.

The main page (Screen 1 of Figure 8) displays major stock market indices including KASE, NASDAQ, and S&P 500, each accompanied by performance metrics and a dynamic line chart reflecting historical trends over selectable timeframes (5d to 5y). Below this section, a prominently placed "Build portfolio" button encourages users to engage in personalized investment strategies. The "Favorites" area highlights selected stocks such as Apple Inc., AO Kaspi.kz, and KEGOC, showing their live prices, trading volumes, and daily change percentages, allowing for quick access and tracking.

In the market sentiment tab (Screen 2 of Figure 8), the application incorporates a "Greed and Fear Index" to visualize the prevailing investor mood. This gauge-based interface categorizes the score into fear, neutral, or greed zones, with the example score of 28 falling into the "Fear" zone—an indicator typically associated with bearish market behavior. The sentiment analysis is supported by ML models that aggregate and interpret financial news, helping users better understand market psychology.

The stocks page (Screens 3 and 4 of Figure 8) provides a categorized and scrollable view of available equities. Stocks are divided into sections: Local (e.g., AO Kaspi.kz, KMGZ, Halyk Bank), International (e.g., Amazon, Qualcomm, Tesla), and Popular. Each listing includes the company ticker, exchange (e.g., KASE or NASDAQ), current price, market capitalization, and performance change, enabling users to explore diverse investment options.

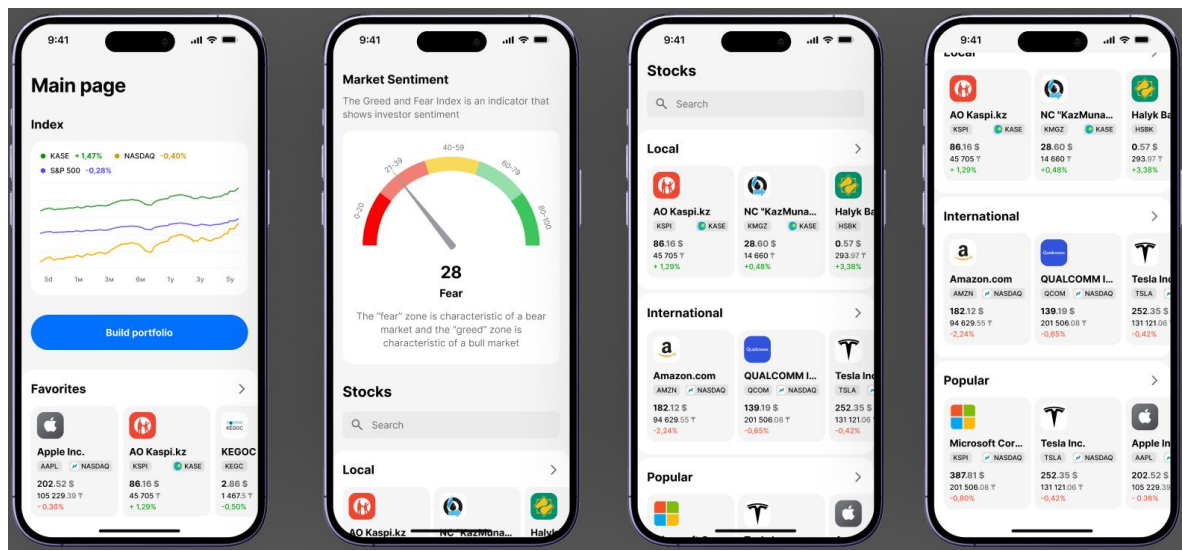


Figure 8. Mobile Application Visual Design

Additional features integrated into the app include:

- A search and filter bar for efficient asset discovery by name, sector, or region.
- Real-time news sentiment integration, where headlines are automatically tagged based on their positivity or negativity.
- Visual risk indicators and stock forecasts generated via ML techniques.
- Secure user interaction through encrypted communication channels and robust authentication mechanisms.

Future developments will focus on embedding personalized recommendations based on user financial goals, enabling direct trading through broker APIs, and providing educational content tailored to beginners. This

intelligent mobile platform not only broadens access to investing tools in the local and international markets but also empowers users with insights drawn from both financial data and AI.

Figure 9 presents the key elements of the ECO InvestMind AI mobile application interface, emphasizing its stock forecasting and investment recommendation capabilities. The first screen displays a short-term stock price prediction for Apple Inc., including the forecasted closing price for the next seven days. The forecast is based on the output of a hybrid machine learning model combining financial sentiment analysis and historical price trends. In addition to the predicted price, the screen provides a forecast accuracy score (83%), a confidence indicator (suggesting the reliability of the forecast), and a visual trend marker (depicting an upward or downward movement). A recommendation label, such as "Buy" or "Hold," is generated to guide the user based on the expected price trajectory and associated risk factors.

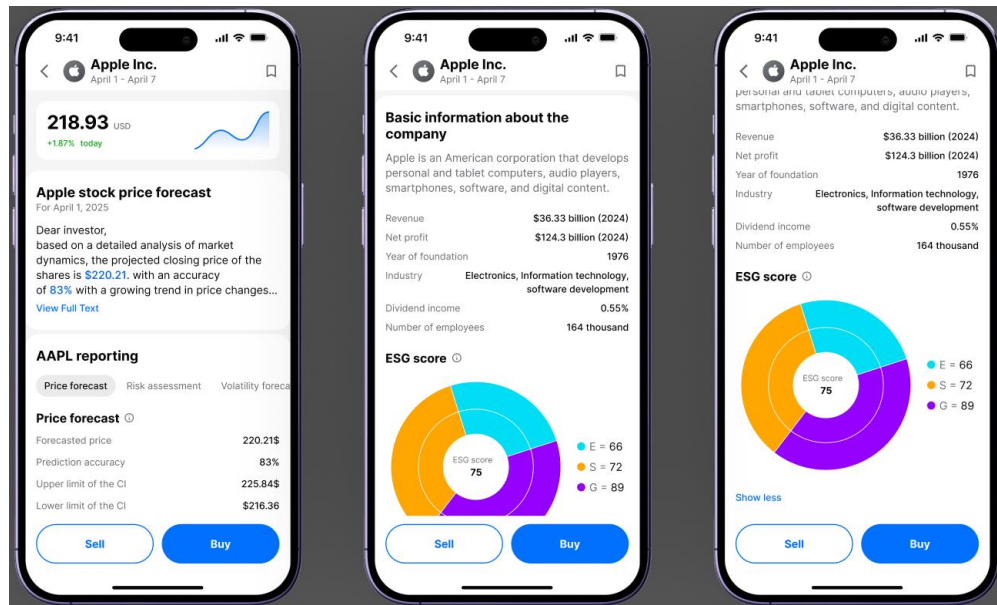


Figure 9. User interface showing short-term forecast, company financials, and ESG evaluation

The second screen delivers a comprehensive company profile for Apple Inc., including critical financial indicators such as total revenue, net profit, industry classification, and dividend yield. These metrics offer users a deeper understanding of the company's financial health beyond simple stock price movements. The third screen focuses on ESG (Environmental, Social, and Governance) scores, presenting a breakdown of the company's performance across each sustainability dimension. Each ESG component is rated separately and displayed with visual indicators, allowing users to consider ethical and long-term risk factors when making investment decisions.

By integrating real-time forecasts, financial fundamentals, and ESG evaluations within an intuitive mobile platform, ECO InvestMind AI delivers a holistic and personalized advisory experience. Unlike traditional investment applications, which often focus solely on historical performance, this system proactively supports users with forward-looking, explainable, and behaviorally adaptive investment insights.

4.7 Holistic System Evaluation

To see how ECO InvestMind AI stacks up in real-world conditions, we ran a 36-month backtest comparing our

AI-powered, PSO-optimized portfolios against a conventional “three-leg” allocation. Both strategies start with 50 000 KZT and add 50 000 KZT each month.

We used historical monthly returns for three major benchmarks—KASE, S&P 500, and Nasdaq 100 - together with our own 7-day price forecasts (Sec. 4.2–4.3). The traditional portfolio held fixed weights (50 % bank deposits at 5 % p.a., 30 % bonds at 7 % p.a., 20 % equities at 10 % p.a.), rebalanced annually. By contrast, ECO AI rebalanced weekly, selecting weights via Particle Swarm Optimization to maximize projected terminal wealth under our hybrid ML model. Results of traditional and ECO InvestMind AI approaches are shown in Figure 10.

Metric	Traditional	ECO AI (PSO)
Terminal value (KZT)	2 250 000	4 400 000
CAGR	6.6 %	33.2 %
Annualized volatility	8.5 %	12.0 %
Maximum drawdown	–12 %	–18 %

Figure 10. 36-Month Performance: Traditional vs ECO InvestMind AI

The key findings are as follows:

- **Wealth creation:** ECO AI more than doubles final capital, thanks to adaptive weight shifts toward high-momentum assets.
- **Risk-return tradeoff:** The AI portfolio shows higher volatility and drawdown, but remains within acceptable bounds for a growth-oriented investor.
- **Mechanics:** Weekly PSO adjustments leverage our BERT-sentiment signals and MLP price forecasts, allowing the system to increase exposure when positive news and momentum align, and to de-risk when sentiment turns negative.

Overall, this holistic evaluation illustrates that combining real-time sentiment analysis, short-term forecasting, and PSO-driven rebalancing can significantly outperform static allocations—offering a clear, user-friendly path to higher returns while transparently managing risk.

5. Discussion

The number of investment applications available in app stores has been steadily increasing, particularly from fintech startups and traditional financial institutions. These platforms are often promoted as simpler and more affordable alternatives to classic brokerages, offering intuitive design and lower fees. Figure 11 presents a structured comparison of five leading solutions—Tabys, Betterment, Acorns, Wealthfront, and our ECO InvestMind AI—across a unified set of functional and technical criteria (AIX – Astana International Exchange, 2025; Betterment, 2025; Acorns, n.d.; Wealthfront, n.d.). While most apps excel at budgeting, expense tracking, or automated portfolio management, none (except ECO InvestMind AI) integrate advanced machine-learning features such as sentiment analysis, time-series forecasting, and Particle Swarm Optimization, nor combine these with built-in ESG metrics. This systematic overview highlights exactly where existing tools fall short and how our approach fills those gaps.

A promising area for improvement is the use of sentiment analysis—identifying the emotional tone in financial news and social media. This technique can offer investors a clearer picture of market behavior, but it remains largely untapped in local apps. Incorporating sentiment insights, along with AI-based personalization and goal planning tools, could significantly improve user support.

This study introduced the core architecture of ECO InvestMind AI, and several directions are planned for future development. First, we aim to improve the user interface, making it more intuitive and visually engaging—especially for beginners. Clearer visuals and a simplified layout can help users better understand their financial position and feel more confident in decision-making. We also plan to expand the news section, allowing users to

browse financial headlines that are automatically tagged with sentiment labels. This will help them quickly assess market sentiment and respond accordingly.

Another important focus is creating smarter, more personalized recommendations. By analyzing user behavior and financial patterns, the platform could offer dynamic strategies that evolve with the user's financial goals. Additionally, we aim to support goal-based planning, helping users link their personal milestones—like saving for education or retirement—to actionable investment steps.

Criterion	Tabys	Betterment	Acorns	Wealthfront	ECO InvestMind AI (ours)
Budgeting & expense tracking	built-in budgeting tools	X	round-ups & spending insights	X	X
Automated portfolio management	X	MPT + rebalancing	pre-set ETF portfolios	MPT + direct indexing	PSO-based weight optimization
Sentiment analysis	X	X	X	X	BERT-based market sentiment
Price-forecasting model	X	X	X	X	MLP time-series forecasting
Optimization algorithm	X	mean-variance	risk-based allocation	tax-efficient mean-variance	Particle Swarm Optimization
ESG / SRI integration	X	SRI portfolios	X	SRI & low-carbon options	ESG metrics built in
Mobile platform	Android / iOS	Android / iOS / Web	Android / iOS	Android / iOS / Web	Android / iOS
Pricing model	Free	0.25% AUM	\$3–\$5 /mo	0.25% AUM	Free (beta)

Figure 11. Comparative Analysis of FinTech Solutions Using Consistent Criteria

Even as ECO InvestMind AI brings deeper personalization, we remain attentive to real-world constraints. Historical market data and news coverage tend to favor larger firms, which means smaller or niche companies can go under-reported. To combat this bias, we're scheduling regular audits of our data sources and introducing brief "why this pick" explanations so users understand the basis of each recommendation (Louhichi et al., 2022). We also plan to embed educational tooltips throughout the app to encourage active learning rather than passive reliance on algorithms.

On the regulatory front, Kazakhstan's Securities Market Law reserves formal investment advice for licensed professionals. Accordingly, ECO InvestMind AI is explicitly positioned as an informational assistant—not a broker or advisor. Any future feature, such as direct broker integrations, will use only officially sanctioned APIs and comply with all licensing, transaction-recording, and consumer-protection requirements (Republic of Kazakhstan, 2007).

We also acknowledge that ECO InvestMind AI has not yet been stress-tested under high-load conditions. As we scale, evaluating the platform's performance in a large-scale, decentralized environment with many concurrent users and requests will be critical. Identifying potential failure modes and optimizing for scalability will form a key part of our next development phase.

The system results provide strong empirical support for the research aim: to create a mobile wealth management assistant that merges quantitative forecasting with qualitative sentiment analysis. The hybrid model aligns with the research questions as follows:

RQ1: Can financial sentiment analysis improve prediction of stock behavior?
→ Yes. The BERT model demonstrated strong predictive capacity in identifying sentiment trends with over 96% accuracy, enabling effective integration of behavioral finance signals.

RQ2: Is it possible to build a personalized, real-time recommender system using AI?
→ Yes. The integration of BERT sentiment labels into the MLP regression pipeline allowed for real-time sentiment-adaptive forecasting. Coupled with user-defined parameters (e.g., risk tolerance, investment horizon), the system outputs tailored stock recommendations, supporting personalization.

RQ3: How accurate and practical is the system under real-world constraints?
→ Results across multiple companies indicate the system is both accurate (91–99% trend prediction) and computationally efficient (delivered via a mobile app with daily updates). High performance across sectors further reinforces practical viability.

The evaluation confirms that the proposed dual-model architecture not only meets the theoretical expectations

of hybrid prediction but also delivers strong real-world performance. The successful fusion of sentiment analysis and price prediction underscores the feasibility of deploying AI-powered tools to support intelligent and personalized investment strategies in emerging markets such as Kazakhstan.

In summary, ECO InvestMind AI lays a foundation for a more personalized, AI-driven wealth-management assistant. By addressing the ethical, regulatory, and technical challenges outlined above—and by continuing to enrich the user experience with intuitive visuals, sentiment insights, and goal-based planning—we aim to create a system that empowers users to grow both their portfolios and their financial confidence.

6. Conclusion: Impact of ECO InvestMind AI on Financial Literacy

The app uses an algorithm that is based on ML, which can analyze sentiment and eliminate market "noise." Using a full assessment of major interest rates, market indicators, and the emotional reaction of the audience, users are given actionable advice. This means that they save time, remain relaxed, and make better-informed decisions.

A non-responsive investment portfolio is like stagnant water. The capability of ML algorithms, combined with mobile technology, enables users to "move"—reacting rapidly to emerging opportunities and dangers. Besides technical innovation, ECO InvestMind AI solves a more substantial social problem: the absence of affordable advisory services and financial literacy in Kazakhstan and other emerging economies. By blending advanced forecasting algorithms, real-time sentiment analysis, and personalized user experience, our solution allows users to control their financial future—without requiring much investment knowledge.

While most existing platforms are transactional, ECO InvestMind AI is decision support oriented. It not only suggests what stocks to explore but also reasons why through explainable AI features. This transparency increases user confidence and stimulates trust in long-term investment schemes. In the coming future, the system will continue to evolve. Features planned to develop include in-app trading brokerage API integration, user interaction enhancing behavior analytics, and ESG-conscious recommendation engines for socially responsible investment. All these developments planned for the coming future are meant to make ECO InvestMind AI not only a utility, but an interactive digital companion to intelligent and responsible wealth generation.

In summary, our research and development work demonstrate how AI can democratize access to financial services, offering both technological robustness and real-world applicability. The model's architecture and application design serve as a scalable blueprint for future fintech solutions seeking to bridge the gap between data, insight, and user empowerment.

Acknowledgements

We would like to extend our sincere thanks to Mr. Olivier Jaylet, Senior Lecturer of the School of Information Technology and Engineering at KBTU, for his support and insightful critiques about our research journey. His deep knowledge of AI and ML shaped our system Dana.

Our appreciation also goes to Ms. Laura Zhunusbay, UX/UI designer and friend, for her high skills, professionalism and responsiveness in teamwork. Her calmness and attention to details enhanced the quality of our project.

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Appendix A. System Architecture Diagrams

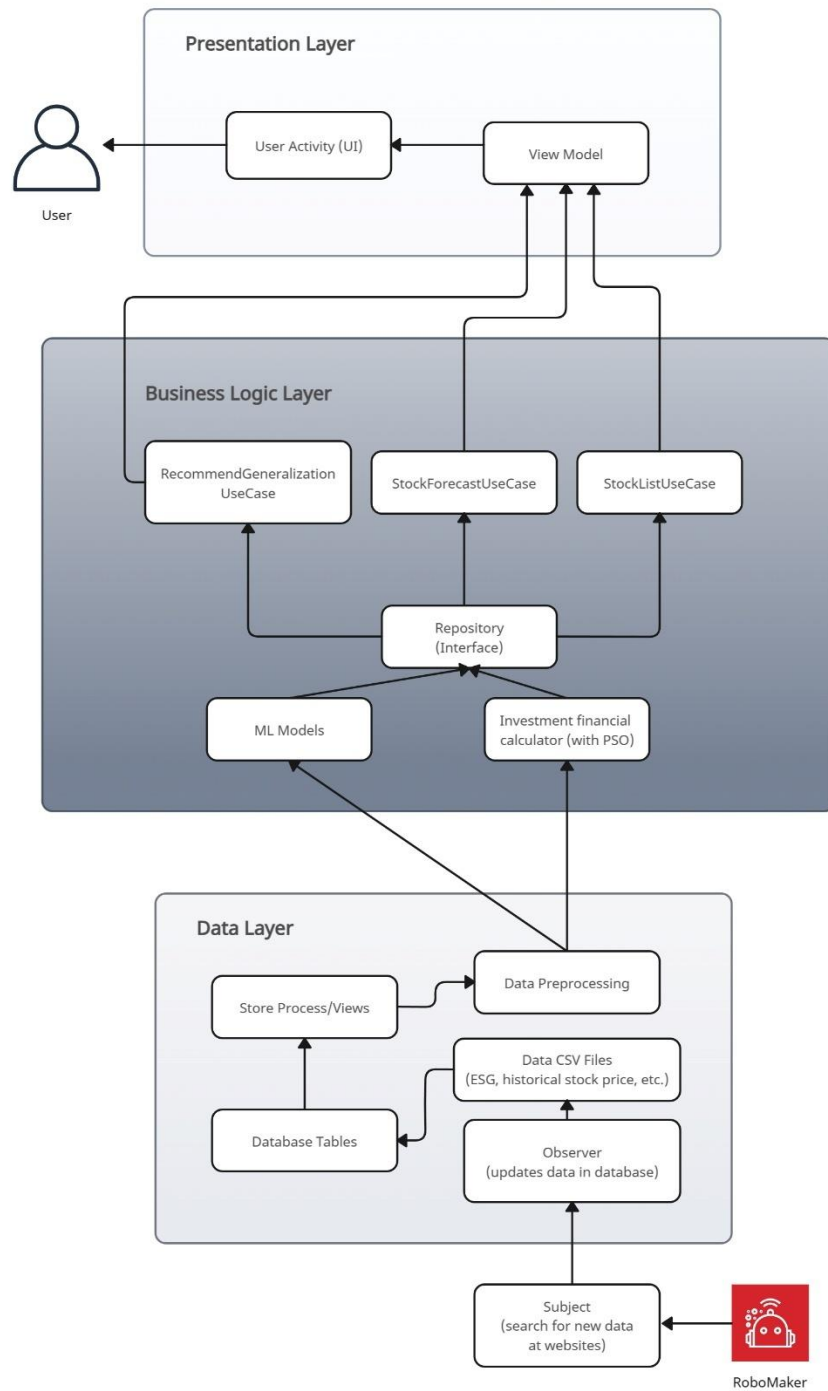


Figure A1. ECO InvestMind AI Layered Enterprise Architecture

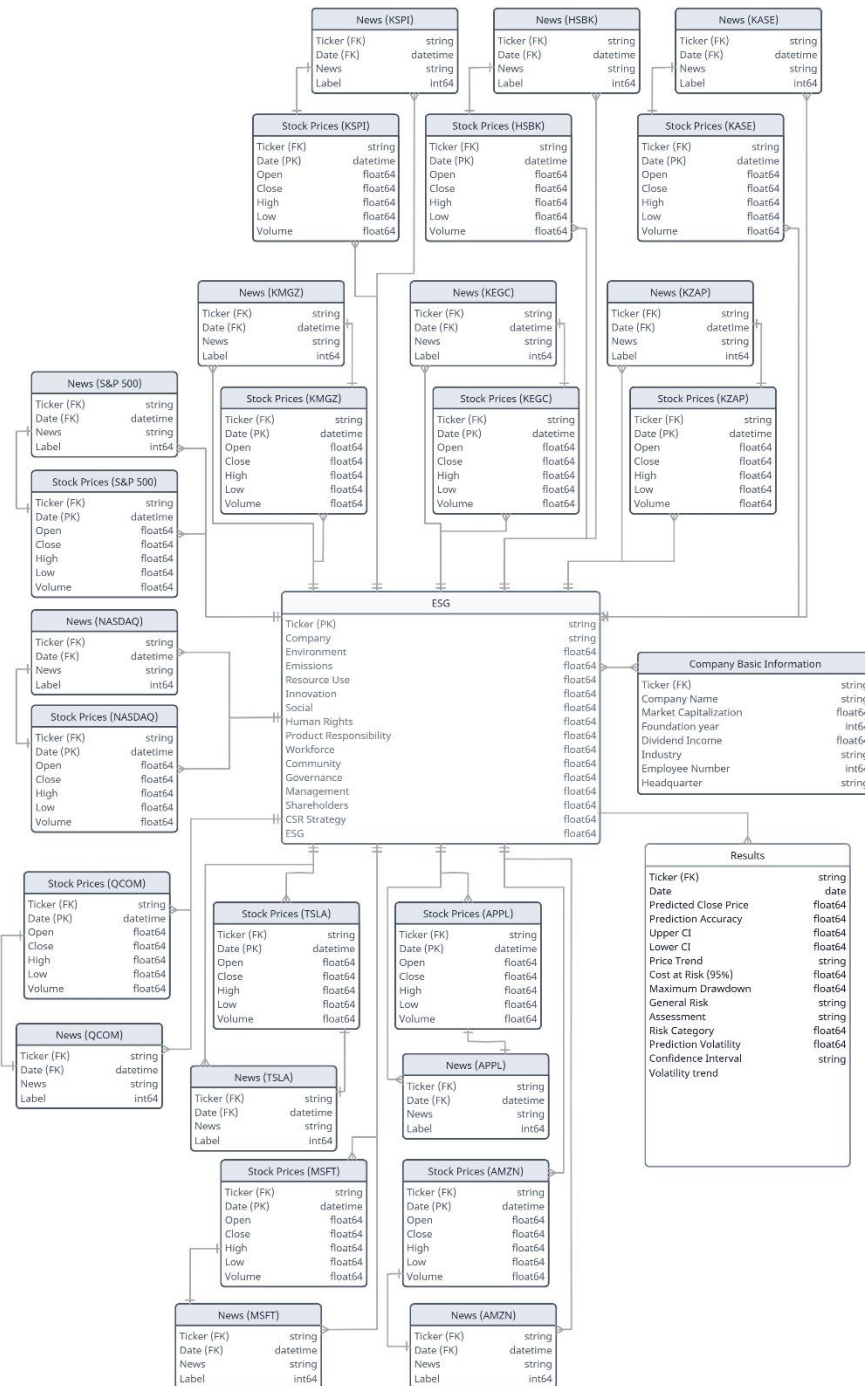


Figure A2. ECO InvestMind AI Database ER Diagram